Optimizing Terminal Passenger Charging Services: A Data-Driven Strategic Blueprint

Memorandum to the Airport Terminal Management Committee

Team GoHiMCM HiMCM 2019 Problem A study pack

Evidence files: `optimal_kiosk_sizing.csv`, `thunderstorm_stress_comparison.csv`

Dear Members of the Committee,

Passengers treat a charged phone as part of the journey product. Dead batteries raise missed-connection risk and desk complaints. We ran a local discrete-event model of charging kiosks for one 16-hour day. We report only measured simulation outputs.

What the model shows

CapEx is capped at \$80,000. Floor area is capped at $40\,\mathrm{m}^2$. A rush-day search evaluated 216 packs. Seven packs remain on the Pareto front. The recommended knee pack is **8 fast kiosks / 16 slow lockers / 0 power-bank banks**. CapEx is \$35,200. That CapEx uses 44% of the CapEx ceiling. Floor use is $38.4\,\mathrm{m}^2$. Rush mean wait is 0.0115 minutes. Rush reneging probability is 0.00.

We injected a thunderstorm delay at 17:00. Forty percent of later departures gained 2–4 hours of delay. Phone drain doubled for delayed travelers. The run has $n = 621$ arrivals. The empirical pack is 12/15/2 at CapEx \$49,000. The knee cut mean wait from 4.591 to 2.347 minutes. That is a 48.9% wait cut. Reneging probability fell from 0.0229 to 0.0114. That is a 50.3% reneging cut. CapEx fell by 28.2%. Peak wait-queue length stayed at 6 for both packs.

Standard operating procedures (SOP)

1. **Three-minute walk grid.** Place charging kiosks so every gate cluster is within a three-minute walk. Prefer corridor nodes that serve two gates. Do not stack all fast kiosks at one retail island.

2. **Hardware mix for Phase 1.** Procure pack 8/16/0 first. Keep \$44,800 of CapEx headroom under the \$80,000 cap. Use headroom only after a second storm-seed review.

3. **Free first 15 minutes on the fast kiosk.** Post a clear tariff: first 15 minutes free on the fast kiosk; meter thereafter. The free window raises turnover and cuts abandoned waits. Keep slow lockers as the long-dwell product.

4. **Storm playbook after** 17:00**.** When Ops declares a weather delay, open a temporary phone-charging lane at the largest hold room. Staff one attendant to load-balance fast kiosks and slow lockers. Publish queue status on gate screens every 10 minutes.

5. **Power-bank policy.** Phase 1 does not require power-bank banks at the knee pack. If retail demands banks, add them only where gate dwell exceeds 90 minutes and CapEx remains under the cap.

6. **Monthly KPI board.** Track mean wait, reneging probability, and peak queue length. Trigger a re-size review if reneging probability exceeds 5% for two consecutive peak weeks.

Ask of the Committee

Approve Phase 1 procurement of pack 8/16/0. Authorize the free-15-minute fast-kiosk tariff pilot for 90 days. Direct Ops to draft the 17:00 storm lane checklist before the next severe-weather season.

We can supply the locked CSV tables and figures on request. All figures are 300 DPI outputs from the same local Python runs.

Respectfully,

GoHiMCM Study Team
Airport Charging Resilience Workstream